



AZIV AI

FINAL MASTER SOFTWARE BLUEPRINT

Laravel / PHP • Universal AI • Fully Admin-Controlled SaaS

Prepared from the original Aziv AI Complete Software Blueprint, revised to make the application PHP/Laravel-hosting friendly and to maximize Admin Panel control, theme control, provider/model flexibility, and future scalability.

Core principle: As many operational, branding, AI, billing, content, security and feature settings as reasonably possible must be configurable from the Admin Panel without routine code changes.

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1. PROJECT VISION

- Aziv AI will be a centralized multi-AI SaaS platform where users can access multiple AI providers and capabilities through one application.
- The original blueprint's core flow remains: authentication → subscription/credit check → cache/context → AI routing → provider → response → usage/cost logging → storage → user response.
- The application will use PHP/Laravel as the primary backend/application framework for practical deployment on Laravel/PHP hosting.
- The system must be modular, secure, API-driven, scalable and heavily Admin-configurable.

2. TECHNOLOGY STACK — FINAL

- Backend/Application: PHP 8.3+ where supported and the latest stable Laravel version compatible with the selected hosting environment.
- Frontend: Laravel Blade + Livewire + Alpine.js, with Tailwind CSS or an equivalent design-token system.
- Database: MySQL 8+ for broad hosting compatibility; PostgreSQL may be used for larger deployments.
- Cache/queues: Redis when available; database/file alternatives for basic hosting.
- Storage: local storage for MVP where appropriate, with S3-compatible object storage for scale.
- The architecture must avoid unnecessary dependence on Node.js/Next.js for the core application.

3. ADMIN PANEL — MASTER CONTROL CENTER

- The Admin Panel is the operational control center of Aziv AI. Administrators should be able to configure the platform without routine code changes.
- All settings must have clear labels, validation, safe defaults, permission controls and audit logging.
- Sensitive settings must be masked and accessible only to authorized roles.
- Where a setting requires deployment-level infrastructure rather than application configuration, the Admin Panel should clearly indicate that limitation rather than pretending it can control it.

4. BRANDING & LOGO CONTROL

- Upload/change primary logo, dark/light logo, compact logo, login logo and email logo.
- Upload/change favicon and app icons.
- Change company/product name, short name, website title and tagline.
- Change support email, support phone, website URL and social/contact links.
- Change footer copyright text and legal/company information.
- Upload default user avatar and system placeholder assets.
- Enable/disable visible branding elements by page where practical.
- Provide media library with upload, replace, preview and safe deletion controls.

5. MULTI-THEME & COMPLETE COLOR CONTROL

- Multiple selectable built-in themes, for example: Light, Dark, Midnight, Professional, Minimal, Glass, Ocean and Custom.
- Admin can create, duplicate, rename, activate/deactivate and delete custom themes.
- Each theme uses design tokens rather than hard-coded page colors.
- Admin controls: primary, secondary, accent, background, surface, card, border, text, muted text, success, warning, danger, info, links, hover, active and focus colors.
- Controls for sidebar, navbar, buttons, inputs, cards, tables, badges, modals, chat bubbles and code blocks.
- Separate light and dark palettes.
- Optional custom CSS field with strict permission and sanitization controls.
- Preview theme before publishing and restore the previous active theme.
- Allow per-brand configuration such as border radius, shadow intensity, spacing scale and typography choices.

6. UI / UX CONTROL

- Admin can control sidebar menu labels, visibility and ordering where practical.
- Control dashboard widgets, landing-page sections, banners and announcements.

- Control pagination defaults, table density and date/time display settings.
- Control language/locale defaults and available UI languages.
- Control timezone and currency defaults.
- Control maintenance mode, maintenance message and maintenance page.
- Control user-facing feature visibility through feature flags.

7. HOMEPAGE & CONTENT MANAGEMENT

- Hero title, subtitle, CTA labels and links.
- Feature sections, benefit sections, model/provider sections, pricing sections and FAQ.
- Banner/announcement creation, scheduling, priority and active status.
- Footer links, legal links, privacy and terms links.
- SEO title, description, keywords where applicable, social preview metadata and favicon.
- Contact/support content.
- All editable content should be validated and safely rendered.

8. USER & ACCOUNT MANAGEMENT

- View/search/filter users and user status.
- Activate, suspend, verify or restrict accounts according to role permissions.
- View subscription, credits, usage summary and account metadata.
- Change plan and adjust credits with reason logging.
- Configure registration availability, email verification requirement and password policy.
- Configure optional Google/OAuth login when credentials are available.
- Configure user session limits, idle timeout and rate limits.

9. ROLES & PERMISSIONS — GRANULAR

- Super Admin, Admin, Support Manager, Finance Manager and Content Manager remain suggested roles.
- Add custom roles from Admin Panel.
- Create granular permissions for users, providers, credentials, models, routing, billing, credits, content, themes, analytics, security and logs.
- Support deny-by-default for sensitive modules.
- A support role must not automatically gain access to API credentials or financial configuration.
- Record role and permission changes in activity logs.

10. UNIVERSAL AI PROVIDER MANAGER

- Admin can add, edit, disable and test AI providers.
- Provider fields: name, slug, API base URL, API format, authentication method, credential reference, status, priority, region, supported capabilities and health settings.
- Support official/provider-specific adapters and a generic OpenAI-compatible adapter.
- Support custom HTTP/API mappings for compatible documented APIs.
- Provider credentials remain encrypted/server-side and never appear in frontend source.
- Multiple credentials can be stored for legitimate rotation/redundancy, subject to provider terms.
- Admin can set provider budgets, rate limits, fallback provider and maintenance status.

11. AI MODEL CATALOG & AUTO-SYNC

- Do not hard-code a permanent 'latest model' into business logic.
- Where a provider exposes model discovery, synchronize its available models.
- Admin can manually add/edit models when automatic discovery is unavailable.
- Model fields: provider, model ID, display name, status, modality, context size, max output, reasoning/tool/vision/audio/image capability flags and pricing metadata.
- Statuses: Stable, Preview, Experimental, Deprecated and Disabled.
- Admin can enable/disable a model globally or by subscription plan.
- Provide 'Refresh Models' and scheduled model synchronization.
- Log synchronization results, failures and model changes.

12. PROVIDERS TO SUPPORT / EXTEND

- Initial architecture: OpenAI and Google Gemini.
- Recommended adapters: Anthropic Claude, DeepSeek, Mistral and Groq.
- Optional aggregators/providers: OpenRouter and Hugging Face Inference Providers.
- Future adapter-ready providers may include xAI, Cohere, Perplexity, Together AI, Replicate, Fireworks, Cerebras and other approved APIs.
- The system must not claim that every internet API is automatically supported. Instead, the Custom Provider and adapter architecture should make future integrations practical.

13. FREE + PAID API MANAGEMENT

- Provider/account classification: Free, Paid, Enterprise/Custom or Unknown.
- Model billing classification: Free, pay-per-token, pay-per-request, time-based, subscription-included or custom.
- Track actual provider cost separately from the price/credits charged to Aziv AI customers.
- Admin can configure credit costs per model/capability and promotional/free allowances.
- Free-tier usage must still respect provider quotas, rate limits and terms.
- Never use multiple keys to circumvent provider quotas, billing, rate limits or restrictions.

14. SMART AI ROUTER

- Auto routing considers request type, user plan, available credits, model capability, provider health, latency, cost, budget, priority and fallback rules.
- Routing modes: Auto, Best Quality, Fastest, Lowest Cost, Free Only, Admin Preferred, Specific Provider and Specific Model.
- Capability-aware routing prevents a text-only fallback from being used for a vision/image/audio request.
- Transient errors may use controlled retry with exponential backoff.
- Repeated provider failures trigger a circuit-breaker/temporary unhealthy state.
- Every routing decision is logged for diagnostics and cost analytics.

15. CHAT SYSTEM

- New chat, chat history, search, rename, delete, streaming, stop generation, regenerate, copy and feedback.
- Manual model selection and Auto mode.
- Conversation context management with configurable limits.
- Optional system prompt/persona configuration controlled by Admin.
- Admin-configurable maximum message length, attachment limits, rate limits and plan-specific usage rules.
- Conversation retention and deletion rules must respect privacy requirements.

16. IMAGE GENERATION

- Text-to-image generation with provider/model selection.
- Image credits, generation history, status tracking, prompt history and regeneration.
- Admin controls allowed image providers/models, credit cost, limits, file size, retention and access.
- Architecture ready for image editing, background removal and upscaling.

17. FILE ANALYSIS & KNOWLEDGE BASE

- Upload → validation → storage → extraction → optional malware/security scan → AI context/RAG → response.
- Admin controls allowed file types, maximum size, retention, storage limits and plan access.
- Optional embeddings/vector search module for knowledge bases.
- Admin can create knowledge bases, assign access and configure retrieval settings.
- Sensitive file access must follow role-based permissions and privacy rules.

18. VOICE / AUDIO / FUTURE VIDEO

- Voice input → speech-to-text → Aziv AI → response → text-to-speech.
- Admin controls voice providers/models, quotas, credit costs and feature availability.

- Realtime voice can be added as a separate module.
- Video-ready provider adapter architecture may be introduced later without changing core user/billing architecture.

19. SUBSCRIPTION & CREDIT CONTROL

- FREE, PRO and PREMIUM plans remain fully configurable.
- Admin controls price, currency, billing cycle, messages, token/usage limits, image credits, file limits, storage, models, providers and features.
- Credit ledger records grants, deductions, refunds and manual adjustments with reason.
- Support promotional credits, expiry rules and optional rollover rules.
- Payment webhooks must be idempotent so one payment cannot create duplicate credits.

20. PAYMENT & BILLING

- Subscription purchase, upgrade, downgrade, renewal, cancellation, billing history and payment records.
- Payment provider should be selected based on countries served, business structure, compliance, currencies and available integrations.
- Admin controls plan pricing, coupons/promotions, tax display settings where applicable and invoice settings.
- Never store raw payment card data in the application.

21. COST & PROFIT ANALYTICS

- Daily/monthly/provider/model/user-plan estimated API cost.
- Customer revenue, subscription sales and credit consumption.
- Cost-versus-revenue and estimated margin indicators.
- Provider budget alerts and configurable thresholds.
- Admin can set action rules when a budget threshold is reached: alert, restrict model, route to lower-cost eligible model or disable provider.

22. NOTIFICATION SYSTEM

- In-app and email notifications.
- Admin can create announcements, offers, maintenance notices, subscription reminders and system updates.
- Configure notification templates, subject lines, content, status and target audience.
- Future-ready for push notifications.

23. SECURITY & AUDIT CONTROL

- Password hashing, secure sessions/tokens, CSRF protection, validation, rate limiting and API protection.
- Encrypted provider credentials and least-privilege access.
- Optional MFA for privileged administrators.
- Activity logs for provider, credential, model, routing, subscription, payment, role, theme and user-status changes.
- Backup/recovery strategy, monitoring and alerting.
- Production security testing before launch.

24. ADMIN SYSTEM SETTINGS

- Site status, maintenance mode, default locale, timezone, currency, pagination, upload limits and retention settings.
- Email/SMTTP configuration, notification settings and support configuration.
- Application URLs, storage driver selection and queue configuration where supported.
- Default AI routing mode, default provider priority and default model policy.
- Global usage/rate limits and emergency kill switches for individual providers/features.
- Feature flags for staged releases.

25. API / INTEGRATION TEST CONSOLE

- Admin-only test screen to select provider/model and send a controlled test request.
- Show request status, latency, provider error class, token/usage metadata and estimated cost where available.
- Never reveal the full secret credential.
- Provide connection-test buttons for provider credentials and model synchronization.

26. DATABASE — MASTER MODULES

- users, user_profiles, subscriptions, subscription_plans, payments, payment_transactions, credit_ledger.
- ai_providers, ai_provider_credentials, ai_models, ai_model_capabilities, ai_model_prices, ai_model_sync_logs.
- chat_conversations, chat_messages, message_usage, api_usage_logs, routing_logs, provider_health_logs.
- files, file_chunks, vector/embedding references, image_generations, audio_jobs, notifications.
- themes, theme_tokens, media_assets, content_pages, banners, feature_flags, system_settings.
- admin_users, roles, permissions, role_permissions, activity_logs.

27. ADMIN-CONTROL MATRIX

- Branding: logo, favicon, name, title, tagline, contact, social links.
- Design: themes, colors, typography, buttons, cards, sidebar, dark/light mode, radius, shadows.
- Content: homepage, banners, announcements, FAQs, footer, legal links.
- AI: providers, credentials, models, capabilities, routing, fallbacks, budgets, pricing, health.
- Users: status, plans, credits, limits, roles and permissions.
- Billing: plans, prices, coupons/promotions, payment configuration and invoices.
- Files/media: upload rules, storage, retention and knowledge bases.
- Analytics: user, AI, provider, cost, revenue and performance dashboards.
- Security: sessions, rate limits, MFA, logs, maintenance mode and emergency feature/provider switches.
- System: localization, timezone, currency, email, storage, queue, cache and operational settings.

28. WHAT CANNOT HONESTLY BE PROMISED AS ADMIN-ONLY

- Some infrastructure changes require hosting/VPS/cloud access, deployment or environment-variable changes.
- Some provider credentials or account-level billing settings must be created on the provider's own dashboard.
- Major code-level features or unsupported provider protocols may still require a developer/AI coding session.
- The Admin Panel should expose every practical application-level setting while clearly separating infrastructure-level settings.

29. DEPLOYMENT & OWNERSHIP

- Production domain, hosting, database, storage, AI provider accounts, payment gateway accounts and source-code repository should remain under the business owner's control.
- Use development, staging and production environments.
- Maintain environment configuration documentation, database migrations, deployment documentation and API/integration documentation.
- For high concurrency, streaming, queues and large file processing, move from shared hosting to VPS/cloud infrastructure as demand requires.

30. DEVELOPMENT ORDER

- Phase 1: Laravel foundation, database, authentication, roles and design-token system.
- Phase 2: Admin Panel foundation + branding + themes + content management.
- Phase 3: Universal AI gateway + provider credentials + model catalog.
- Phase 4: OpenAI + Gemini + chat streaming/history.
- Phase 5: Smart routing + health + fallback + cost/usage logging.
- Phase 6: subscriptions + credits + payments.
- Phase 7: Claude + DeepSeek + Mistral + Groq + generic compatible/custom providers.
- Phase 8: files/RAG + image + voice.
- Phase 9: security hardening, testing, backups, monitoring and production deployment.

- Each phase must be tested before moving to the next.

31. FINAL IMPLEMENTATION RULES FOR THE AI CODING AGENT

Rule 1: Use Laravel/PHP as the primary application stack. Do not silently switch the project to Next.js/Node.js.

Rule 2: Treat this document and the original Aziv AI blueprint as the requirements baseline. Do not remove features without explicit approval.

Rule 3: Build module-by-module. Do not generate a huge untested codebase in one step.

Rule 4: Make application-level settings Admin-configurable wherever technically practical.

Rule 5: Do not hard-code a permanent list of 'latest AI models'. Use provider model discovery/synchronization where available.

Rule 6: API keys/secrets must remain server-side and encrypted/securely managed.

Rule 7: Free APIs/free tiers must never be treated as unlimited, and provider limits/terms must not be bypassed.

Rule 8: Every sensitive Admin action must be permission-checked and audit logged.

Rule 9: After each module, run validation/tests and report exactly what was completed, what failed and what remains.

Rule 10: Preserve backwards compatibility and database migrations when adding providers, models or themes.

32. FINAL TARGET

USER → AUTH → PLAN/CREDIT → CAPABILITY CHECK → AI ROUTER → PROVIDER ADAPTER → AI PROVIDER → RESPONSE/STREAM → USAGE/COST → CREDIT LEDGER → STORAGE → USER

ADMIN → BRANDING / THEMES / CONTENT / USERS / PROVIDERS / MODELS / ROUTING / BILLING / CREDITS / FILES / ANALYTICS / SECURITY / SYSTEM SETTINGS

The result should be a scalable Aziv AI SaaS in which the Admin Panel is the central control layer for branding, appearance, content, AI providers, model availability, routing, user access, subscriptions, credits, analytics and operational feature flags.

Original blueprint reference: Aziv AI Complete Software Blueprint & Developer Requirement Specification, 14 pages. The original blueprint specifies the multi-provider architecture, user/admin modules, security, database modules, recommended technology direction, version roadmap and ownership requirements.